21-242 Recitation 12

Linear Transformations — Q2. T(V)= W

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $T(\vec{0}_V) = \vec{0}_W$.

Course-level solution

$$T(\vec{0}_V) = T(\vec{0}_V + \vec{0}_V) = T(\vec{0}_V) + T(\vec{0}_V).$$

Cancellation (Fields Q4) yields $T(\vec{0}_V) = \vec{0}_W$.
Alternatively, $T(\vec{0}_V) = T(0 \cdot \vec{0}_V) = 0 \cdot T(\vec{0}_V) = \vec{0}_W$.